Description

Illustration



Figure caption

1. Decorative ceiling plaster work © Richard Needham at en.wikipedia [CC BY-SA 3.0] 2. Teeth models of gypsum for dental applications © Humusak at Pixalbay [Public domain] 3. Joint fracture of a hand © Condesign at Pixalbay [Public domain]

The material

Plaster of Paris is exactly that: a plaster that, originally, came from Paris, France. It is made by calcining the mineral gypsum, CaSO4.2H2O, at about 180 C, driving off water to give the anhydrite 2CaSO4. H2O. When mixed with water it rehydrates and sets to a hard, white solid. It is used to make molds and casts for ceramics and sculptures, to make pre-cast ornamental plasterwork on ceilings and cornices, and for orthopedic bandages or casts. In medieval and renaissance times gesso (plaster of Paris mixed with glue) was applied to wood panels or canvas as the ground for tempera paintings.

Compositional summary

2CaSO4.H2O, Gypsum

General properties

Density		1.18e3	-	1.7e3	kg/m^3
Price	*	1.09	-	1.56	GBP/kg
Date first used		1730			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✗
Enclosure	✓
Interiors	✓

Services	✖				

Mechanical properties

Young's modulus		4.5	-	8	GPa
Shear modulus	*	2	-	5	GPa
Bulk modulus	*	3	-	5.5	GPa
Bending modulus	*	4.5	-	8	GPa
Poisson's ratio		0.25	-	0.3	
Yield strength (elastic limit)	*	1	-	4.5	MPa
Tensile strength	*	1	-	4.5	MPa
Compressive strength		14	-	20	MPa
Bending strength		3	-	8	MPa
Elongation	*	0.01	-	0.1	% strain
Hardness - Vickers		1	-	3	HV
Fatigue strength at 10^7 cycles		1.7	-	2	MPa
Fracture toughness		0.008	-	0.012	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.1	-	0.3	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		1.67	-	2.5	m.°C/W
Thermal expansion coefficient		8	-	10	µstrain/°C
Specific heat capacity		600	-	1e3	J/kg.°C
Melting point	*	300	-	500	°C
Maximum service temperature	*	110	-	180	°C
Minimum service temperature		-100	-	-70	°C
Flammability		Highly flammable			
Emissivity		0.9			

Hygro-thermal properties

Frost resistance		Very poor			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	1e6	-	1e8	µohm.cm
Dielectric constant (relative permittivity)		5	-	12	
Dissipation factor (dielectric loss tangent)		0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	2	-	4	MV/m

Optical properties

Natural color

White

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Average			
Sound isolation	Average			

Processability (scale 1 = impractical to 5 = excellent)

Moldability	4	-	5	

Durability

Water (fresh)	Unacceptable
Water (salt)	Unacceptable
Weak acids	Unacceptable
Strong acids	Unacceptable
Weak alkalis	Unacceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Unacceptable
Rural atmosphere	Unacceptable
Marine atmosphere	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.25e8			tonne/yr

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.186	-	0.206	kg/kg
Embodied energy (virgin grade production)	* 2.09	-	2.31	MJ/kg
Water Usage	* 9.79	-	10.8	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 18.1	-	19.9	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.18	-	1.3	kg/kg

Recycling and end of life

Recycle	✗	
Functional recycle	✗	
Recycle fraction in current supply	0.1	%
Combust for energy recovery	✗	
Landfill	✓	
Biodegrade	✗	
A renewable resource?	✗	

Supporting information

Technical notes

In use plaster of Paris is mixed with half its weight of water. It remains usable for 20 minutes, starts to set after 30 and is solid after one hour.

Typical uses

External and internal molded decoration on walls and ceilings; as a mold material for casting low-melting metals; and as medical casts and splints.

Links

ProcessUniverse

Producers

Reference